Configuration Generation and Analysis of AC and DC Conceptual Designs of Shipboard Power Systems

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***Abstract*—In this paper, a generalized network configuration diagram is developed for shipboard power systems. A variety of AC and DC conceptual system designs are developed from the generalized diagram with a list of design constraints implemented. The challenges of these design options in generators, converters, and protection are analyzed and discussed. The benefits and shortcomings of all design options presented in this paper are analyzed and summarized. The design options using different multi-winding generator technologies are especially attractive due to their overall advantages in cost and space saving.**

Keywords*—*shipboard power system, AC, DC, conceptual design, multi-winding generator.

I. Introduction

New equipment and/or new implementations of existing equipment help generate innovative shipboard power system designs. Traditionally, fixed speed AC generators, transformers, and AC breakers forms conventional AC shipboard power systems. The optimum speed operation of variable speed AC generators and the availability of high voltage and high current power electronics converters promote the design and development of DC shipboard power systems. Further, different network configurations facilitate the generation of various design options of AC and DC shipboard power systems.

There is a lack of literature to discuss the generation of different AC and DC shipboard power system design options. In this paper, components are categorized generically. Generalized connection diagrams are developed to connect generators and different types of generic loads using different types of voltage transformation and connection equipment. Constraints are imposed in generating multiple conceptual design options for selecting physically feasible designs. How to develop the generalized diagrams and generate various conceptual system designs are discussed in section II. In section III and IV, various AC and DC system design options are presented. Their main technical challenges, benefits, and shortcomings are discussed. In section V, the features of the presented conceptual designs are summarized, and the generation of more conceptual design options are discussed.

II. Generalized Connection Diagram and Conceptual System Design Generation

A. Generalized Network Configuration

A generalized network configuration diagram for shipboard power systems is illustrated in Fig. 1. For simplification, DC sources, such as energy storages, are not considered in the generalized netowrk connection. Due to the symmetric structure, only the starboard or port side marine electric system is considered in the network configuration analysis.

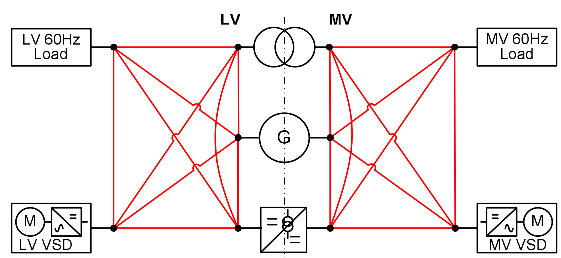


Fig. 1. A generalized AC&DC network configuration diagram for shipboard power systems

Depending on components and connection paths, many conceptual design options can be generated from this genearlized connection diagram. Electric sources can be fixed speed generators or Variable Speed Generators (VSG). There are four types of loads at LV or MV levels with AC or DC connections. Electric power can be delivered from electric sources to the four types of loads with different voltage transomation equipment and/or connection equipment. Three types of voltage transformation equipment are for connecting between MV and LV subsystems. Five types of connection equipment connect various types of components within either MV or LV subsystems. These voltage transformation equipment and connection equipment are summarized in Table I.

TABLE I. VOLTAGE TRANSFORMATION AND CONNECTION EQUIPMENT

Load Types	***MVAC, MVDC, LVAC, LVDC***
Voltage transformation Equipment	1. Generator 2. Transformer 3. DC/DC converter
Connectng equipment	1. AC/DC rectifier 2. DC/AC inverter 3. AC/AC converter 4. Cables or bus duct 5. Transformer

Many paths from sources to loads can be found in Fig. 1. Combining the same type of voltage transformation and/or connection components into one, the generalized network configuration diagrams adopting either a fixed speed generator or a VSG as main source are given as Fig. 2. All branches and connection points are labeled in the figure. Three different types of electric paths, fixed frequency AC, DC, and variable frequency AC, exist. Only one connection equipment is required for connecting most nodes, but two or more equipment in parallel may be needed at special conditions to be discussed later.

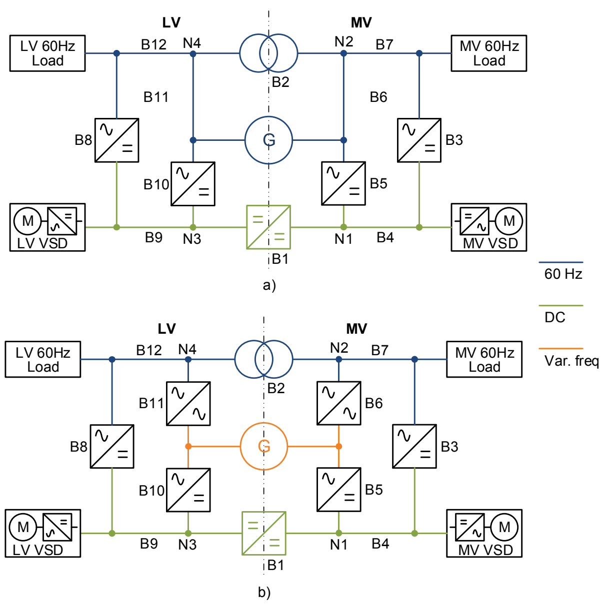


Fig. 2. Generalized network diagram with a) a fixed speed generator and b) a variable speed generator

B. Generation of Conceptual System Designs

Various conceptual system designs can be generated from the generalized connection diagrams in Fig. 2. By removing redundant connections and ensuring power supply to all four loads, multiple AC and DC conceptual system design options can be generated. The generalized diagrams allow an automated method to perform an exhaustive search for all possible conceptual design options.

The generation of a conceptual design is constrained by

- the supply paths to one load. For example, the MV AC and DC loads can only be supplied from the generator's MV output. Therefore, either B5 or B6 should exist in the supply path from the generator to the MV AC or DC load;
- the number of total conversion stages. The maximum number of total conversion stages can be limited. For example, $B_i + B_j + B_k + B_l + B_m + ... \leq N$. N is defined as the maximum allowable conversion stages. B_i ,B_j ,B_k ,B_l ,B_m are the branches connection between the generator and a load, where B_x is equal to one when the branch B_x is connected and zero when disconnected.
- the coupling points of one or more branches. Different coupling points can generate different conceptual designs. Fig. 3 illustrates two different network configuration diagrams having one (N_2) and two (N_1 and N_2) coupling points. The resulted conceptual system designs are given in Fig. 4.

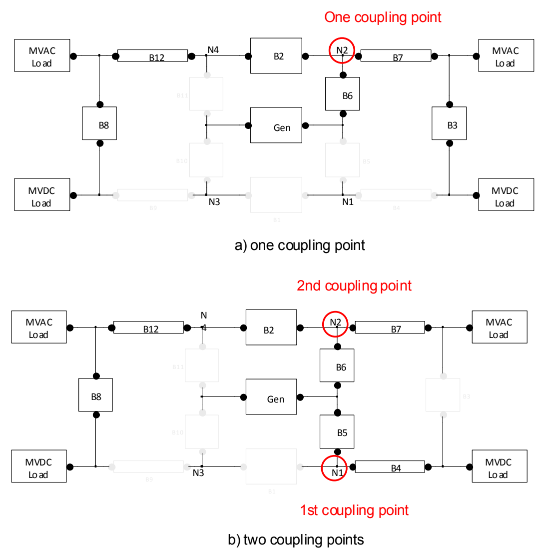


Fig. 3. Two network diagrams with a) one coupling point and b) two coupling points

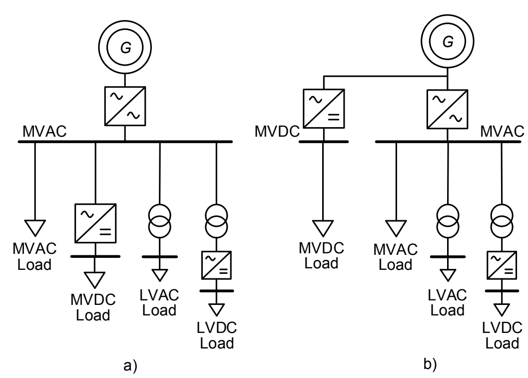


Fig. 4. Two system concepts resulted from Fig. 3 with a) one coupling point and b) two coupling points

III. Representative AC System Conceptual Designs

Several representative AC system conceptual designs can be derived from the two generalized connection diagrams. Beside transformers, multi-winding generators can be implemented for the MV and LV voltage transformation. Each conceptual system design and its major technical challenges are discussed.

A. *Fixed Frequency AC*

Based on Fig. 2a, a conventional 60Hz AC system network configuration diagram is generated as Fig. 5. Two different electric paths, fixed frequency 60Hz AC and DC, exist in this diagram. The corresponding conceptual design diagram is shown in Fig. 6, which is also recognized as the conventional MVAC system. Multiphase MV and LV wye/delta/wye phase shifting transformers are used in the system in order to reduce the harmonics from the converter switching. In this design, the MV 60Hz load is not connected. To connect it, a cable connection in paralle with the transformer is required between the node N_2 and the load.

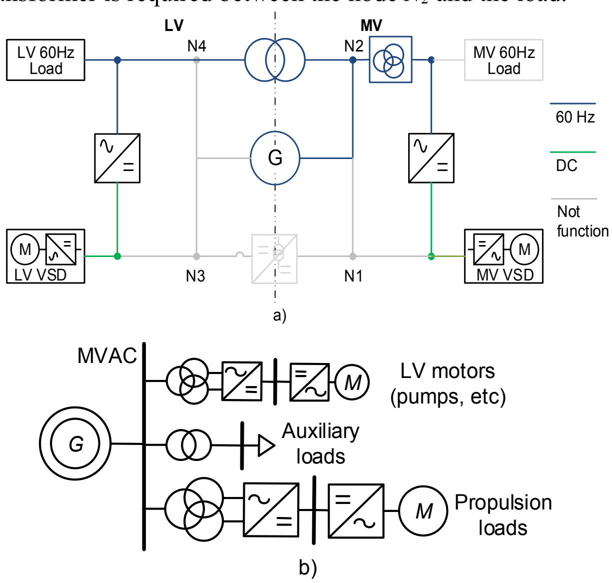


Fig. 5. Fixed frequency AC diagrams a) network configuration b) conceptual system design

B. *Variable Speed Fixed Frequency AC*

A variable speed fixed frequency AC network configuration diagram is developed from Fig. 2b) and shown in Fig. 6a). Three types of electric paths, 60Hz, DC, and variable frequency AC, exist. This variable speed fixed frequency AC system conceptual diagram is illustrated in Fig. 6b). A variable speed generator is connected to a common AC bus through a AC/AC converter. The common AC bus distributes power to all loads. The variable speed fixed frequency system maximizes the utilization of existing AC component and system technologies as well as the generator fuel efficiency by optimum speed operation.

The back-to-back converter is the key to this variable speed fixed frequency conceptual system design. It brings challenges and benefits to the component and system design. Existing AC drives and other applicable converter technlogies should be carefully evaluated and optimized to minimize its overall cost, loss, and footprint. The variable speed generator may be oversized due to harmonics if a low cost diode-front-end back-to-back converter is used. High current THD (Total Harmonic Distortion) induced by the diode rectifier causes saturation and overheating in the generator. Other measures, such as harmonic filters and multiphase generators, may be adopted to reduce harmonics. However, the generator may still be oversized since it should output its nominal voltage even if at its low speed operation. In the conventional AC system design, its capacity is determined by the maximum fault current withstanding of AC circuit breakers. Since the inverter can limit the AC fault currents from the generator, more generators or a higher power installation is allowed in the variable speed fixed frequency design than a conventional AC design.

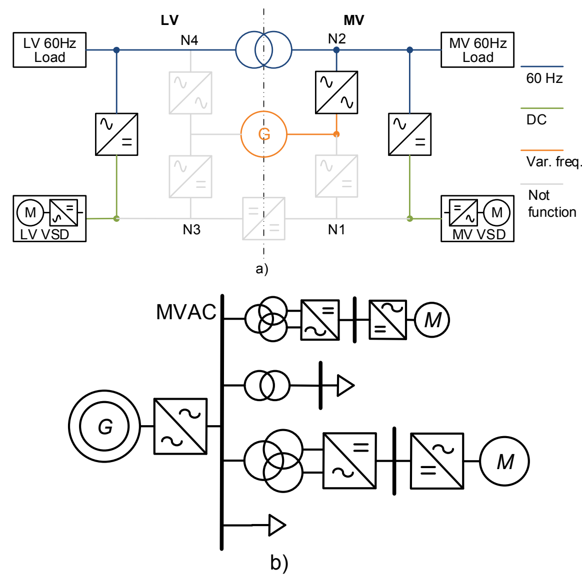


Fig. 6. Variable frequency AC diagrams a) network configuration b) conceptual system design

C. *Multi-winding Generator based AC*

Based on Fig. 2b), a multi-winding generator based connection diagram is illustrated in Fig. 7a). Fig. 7b) demonstrates the corresponding MVAC and LVAC conceptual system design diagram. In this design, the generator has multiple isolated MV and LV stator windings with proper phase shift. The multiple winding generator at MV and LV levels elminate the transformers since the generator itself has the voltage transformation function and the phase-shifting design reduces the harmonics inside the generator. In this way, this design may lead to lower cost, higher efficiency, and smaller footprint than the conventional AC system design.

Multi-phase generators outputing power at the same voltage level have been used in power systems for decades. Multi-winding synchronous generators at different voltage levels were developed and proposed for AC shipboard power systems [1]. One technical challenge for this generator is to ensure acceptable current and torque ripples when the MV and LV stator windings are loaded at different levels. One disadvantage of such a MV and LV generator, by preliminary observation, is that the generator stator windings must be rated separately for different MV and LV peak load. In a typical drilling vessel, for example, the installed load demand at MV and LV levels could each take up to 75% of the total installed generation capacity. The total stator windings of the multi-winding generators may need to be rated at 150% of the rated shaft capacity. To reduce the winding oversize, a frantionally rated transformer may be implemented between MVAC and LVAC buses to reduce the total generator winding rating to the rated shaft capacity.

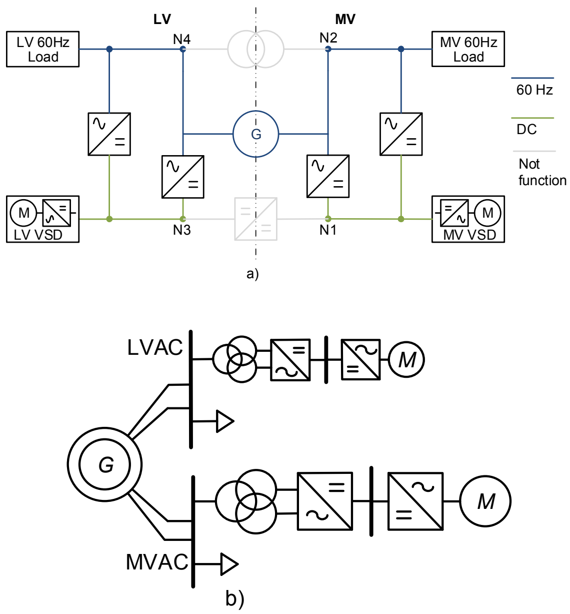


Fig. 7. Multi-winding generator based MVAC & LVAC diagrams a) network configuration b) conceptual system design

Different multi-winding generator technologies may be used for AC shipboard power systems. A DFIG machine delivers rated fixed frequency power from its stator winding and partially rated variable frequency power from its rotor winding. Alternatively, a DFIG based network configuration diagram and its conceptual design diagram are shown in Fig 8a) and b). The DFIG allows higher fuel efficiency than the previous fixed speed multi-winding generators while delivering power to both MVAC and LVAC simulatenously. To save cost, the transformer between MVAC and LVAC may not be needed. Detailed description of this conceptual design can be found in [10].

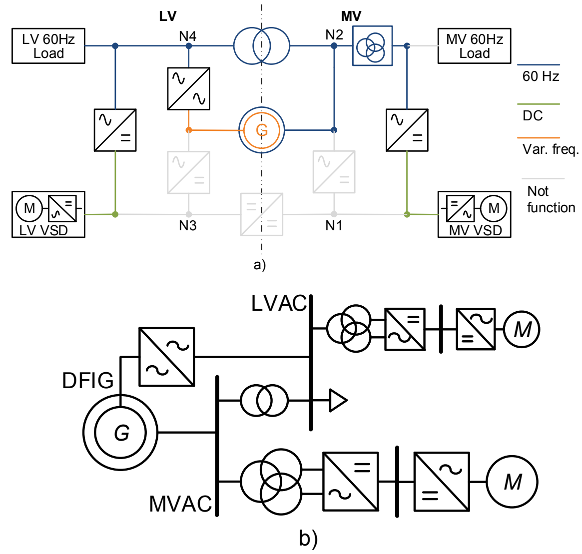


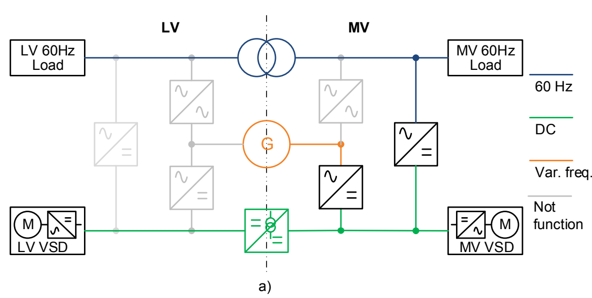
Fig. 8. DFIG based MVAC & LVAC diagrams a) network configuration b) conceptual system design

IV. Representative DC System Conceptual Deisgns And Technical Challenges

Employing VSGs to improve fuel effiency, three representative DC system conceptual designs are derived from the generalized connection diagram in Fig. 2b). The MV and LV voltage transformation uses either DC/DC convereters or generators. Each conceptual system design and its major technical challenges are discussed.

A. DC/DC Converter based DC

A DC/DC converter based DC network configuration diagram can be derived as Fig. 9a) and its corresponding conceptual system design diagram is illustrated in Fig. 9b). The DC/DC converter is responsible for the MVDC and LVDC voltage transformation. All the three types of electric paths, fixed frequency, DC, and variable frequency, exist. A MVDC common bus is to distribute power from the generator to different MV and LV loads. Compared to the conventional AC system, the conversion stages of the MV propulsion load in this MVDC system is one less but those of all other loads increase. Consequently, in certain vessel types, if all other loads are dominant, more conversion stages may result in reduced efficiency and reliability.



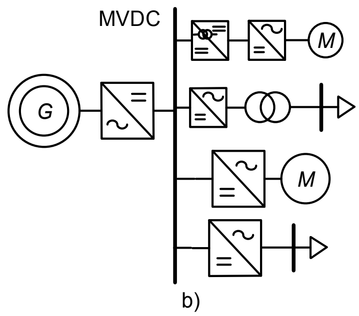


Fig. 9. DC/DC converter based DC diagrams a) network configuration b) conceptual system design

The key components in this conceptual design include the MV generator rectifier, MV inverter, DC/DC converter, and MVDC protection devices. Either Current Source Converters (CSCs) or Voltage Source Converters (VSCs) can be used as the generator rectifier. Among differet CSCs, a thyristor rectifier is perferred for its fault current limiting capability. However, similar to utilizing a diode rectifier in the variable speed fixed frequency AC system, the generator must be oversized for high current THD and over-excitation at its low operation speed. VSCs can reduce current harmonics and maintain rated voltage output even at low operation speed. However, the oversizing because of the extra insulation required by *dv/dt* may still be needed. A Modular Multilevel Converter (MMC) or high-level VSC may replace the conventional two-level VSC since its *dv/dt* is limited and no extra insulation is imposed to the generator design.

MV inverters are needed for interfacing MV propulsion and AC loads in this system. The inverters in existing drives may not be directly applied for the DC system since the harmonics from the inversion operation may be amplified by the resonance in the DC system [11]. The voltage ratio for a DC/DC converter ranges from 5:1 to 20:1. In order to be comparable to the efficiency of a variable speed LV drive system in AC, it is desirable for the DC/DC converter to have an efficiency as high as possible. For the LVAC loads, such as service and hotel loads, AC power supply through a regular transformer is still desired for its high efficiency, safety - galvanic isolation, and low common-mode EMI.

Since all loads are connected to the common DC bus, an cost-effective DC system protection solution is required for practical applications. Due to fast increasing DC fault currents and system wide applications of converters, solid-state devices, either Solid-State Circuit Breakers (SSCBs) or Converters are inherently good for DC fault interruption due to its inherent turning-off speeds. The main challenge of SSCB based MVDC protection solutions is the high cost of the MV semiconductor devices [8]. Most existing converters do not have fault current limiting or interruption capability. The main challenge of converter based MVDC protection solutions is to modify the existing converter topologies and controls in a economical way to allow effective DC fault limiting or interruption [9].

B. Multi-Winding Generator based DC

A multi-winding generator based DC connection diagram can be developed as illustrated in Fig. 10a) and its conceptual system design diagram is demonstrates in Fig. 10b). This design has the major voltage transformation function embedded into the generators, instead of the DC/DC converter. Similar to the previous mult-winding generator based AC designs, this multi-winding generator based DC design may lead to further improvements in cost , efficiency, and footprint than the previous DC/DC based DC conceptual system design.

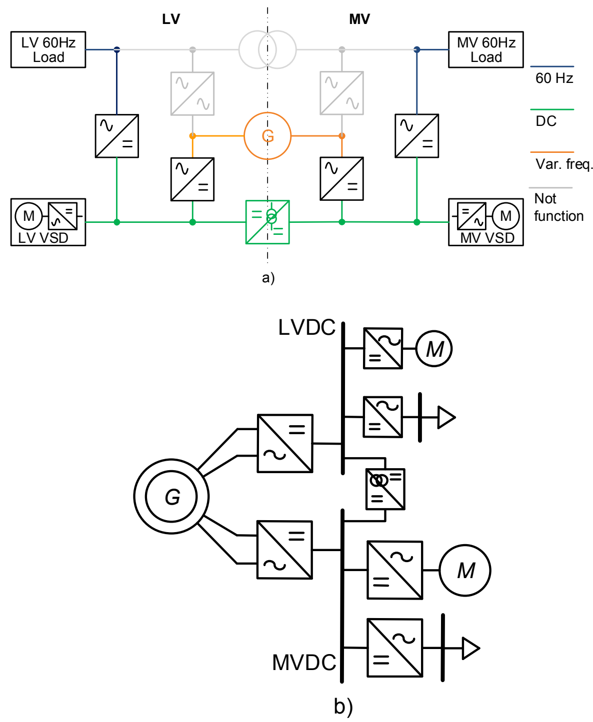


Fig. 10. Multi-winding generator based DC diagrams a) network configuration b) conceptual system design

With VSGs allowed in DC systems, more generator technologies are applicable to this multi-winding based DC conceptual system design. Besides the aforementioned multi-winding synchronous generator and DFIG (Doubly-Fed Induction Generator), open winding synchronous generators [2] and multi-winding induction generator [3] may also be applied for this conceptual system design. With the multi-winding synchronous generator, to reduce the winding oversize caused by the individual sizing of the MV and LV windings, a DC/DC converter with bidirectional power flow capability may be used as in Fig. 8 and is fractionally rated in order to reduce the total generator winding rating to the rated shaft capacity.

A multi-winding generator based hybrid AC and DC connection and its conceptual design diagram are illustrated in Fig. 11. This conceptual design can deliver power to both AC and DC subsystems simulateneously. Since the AC and DC subsystems are coupled at the generator, the mutual impacts between the AC and DC operations should be carefully considered in the generator design. Since the operations of AC and DC are in different time frames, this mutual impacts is more challenging than in a pure AC or DC system. Technical challenges and solutions in the multi-winding generator design, such as harmonics, DC side power oscillations, and DC short circuit, have been discussed in [4]-[7].

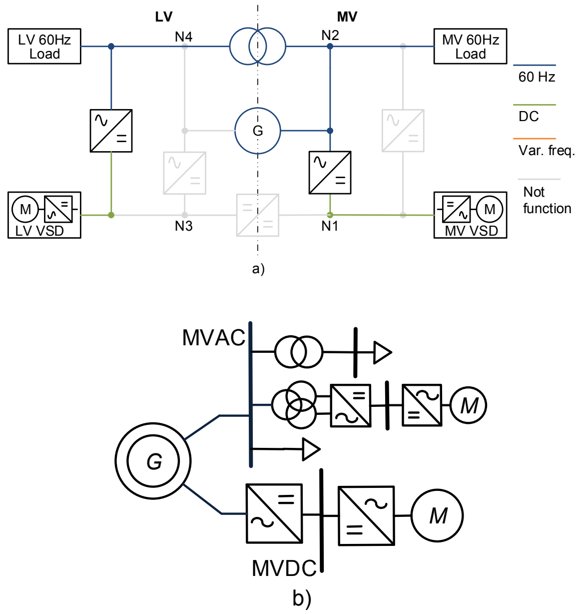


Fig. 11. A multi-winding based hybrid AC&DC a) network diagram b) system concept

V. Summary and Future Work

In this paper, a generic network configuration diagram for shipboard power systems is used to generate several representative AC and DC conceptual system designs. The system cost, reliability, efficiency, size, and volume can be approximated from these conceptual designs. Variable speed generators are better than fixed speed generators for higher fuel efficiency. DC system designs are better than AC system designs in terms of less conversions but have challenges in converter design and DC protection. Various types of multi-winding generators are utilized for coupling different voltage levels and AC and DC. Although main transformers are not removed, the mutual effects between different operations of multiple windings are challenging. The multi-winding generator based conceptual system designs can bring in potential benefits in cost, efficiency, weight and volume.

The generic connection diagram facilitates the generation of novel component and system designs for future shipboard power system development. The representative designs discussed in this paper have the backbone system using conventional fixed frequency AC and DC. Other special conceptual system designs, such as variable frequency AC, variable voltage DC, and variable frequency variable voltage AC designs, can also be obtained from the generic connection diagram. In addition, new and innovative component concepts may also be generated from the newly generated conceptual system designs.

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